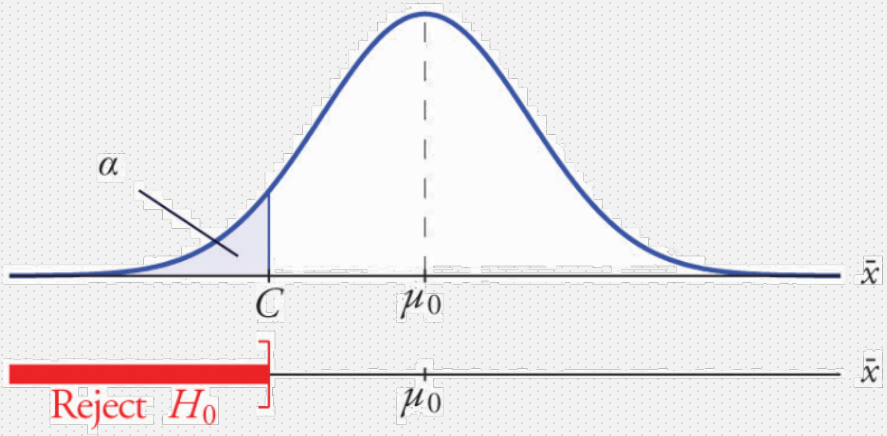
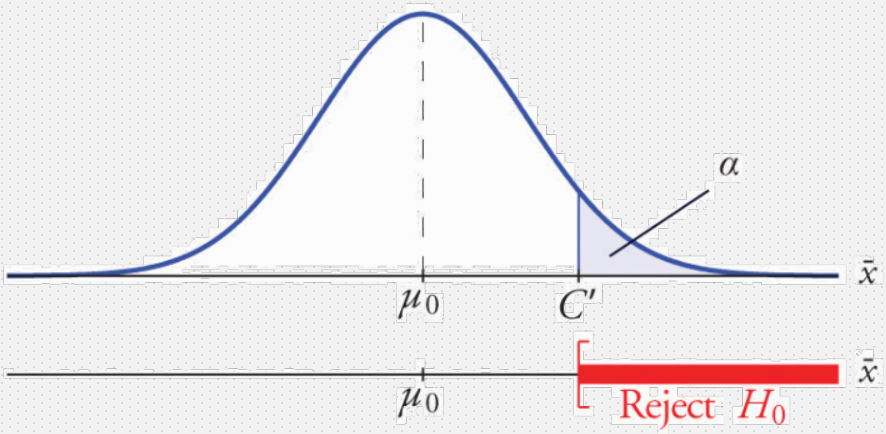
DECEMBER	TOPIC
25	TEST OF HYPOTHESIS (2/2)

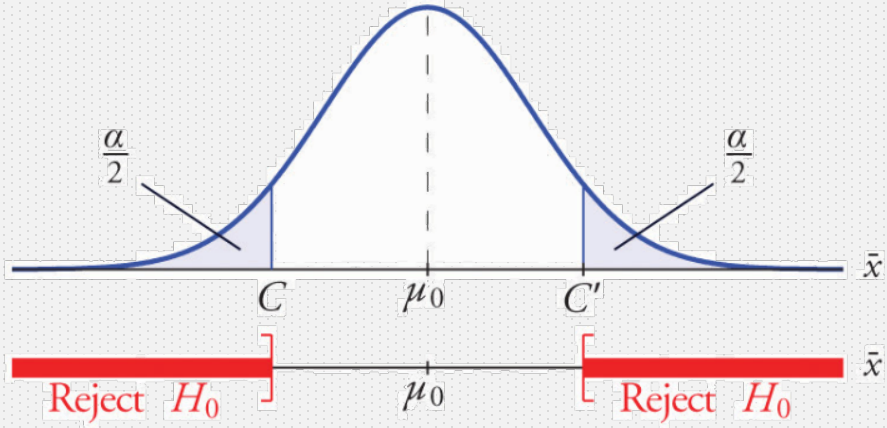
$H_a : \mu < \mu_0$



$H_a : \mu > \mu_0$



$H_a : \mu \neq \mu_0$



EXAMPLE

A geologist claims that the average grade of an iron deposit is greater than 15 %Fe with at the <u>2.5% level of significance</u>.

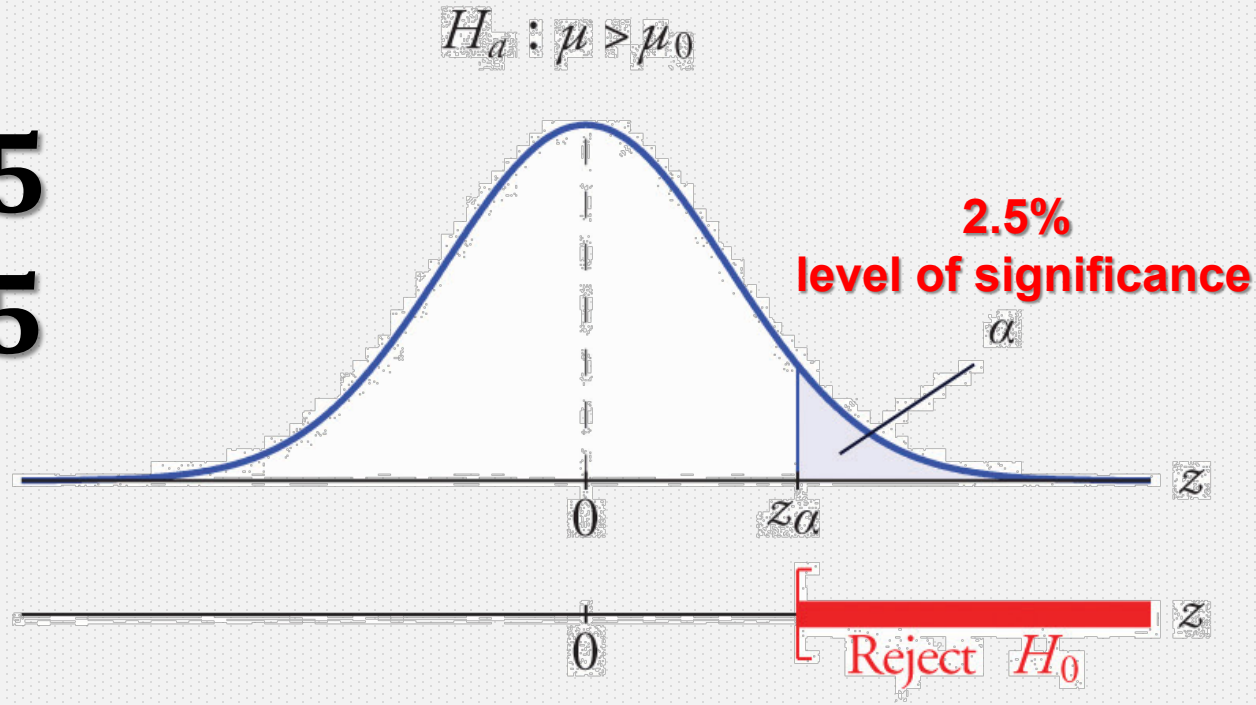
To test this claim, 36 holes are drilled in the deposit and core samples are collected and analyzed for Fe%.

The average and variance of %Fe values are calculated to be 17 Fe% and 9 Fe%. What can we say about whether the claim is true or not ?

EXAMPLE

We want to test if the average grade of the deposit is equal to a certain value. In other words:

$$\mathbf{H_o;\ \mu = 15}$$
$$\mathbf{H_a;\ \mu > 15}$$

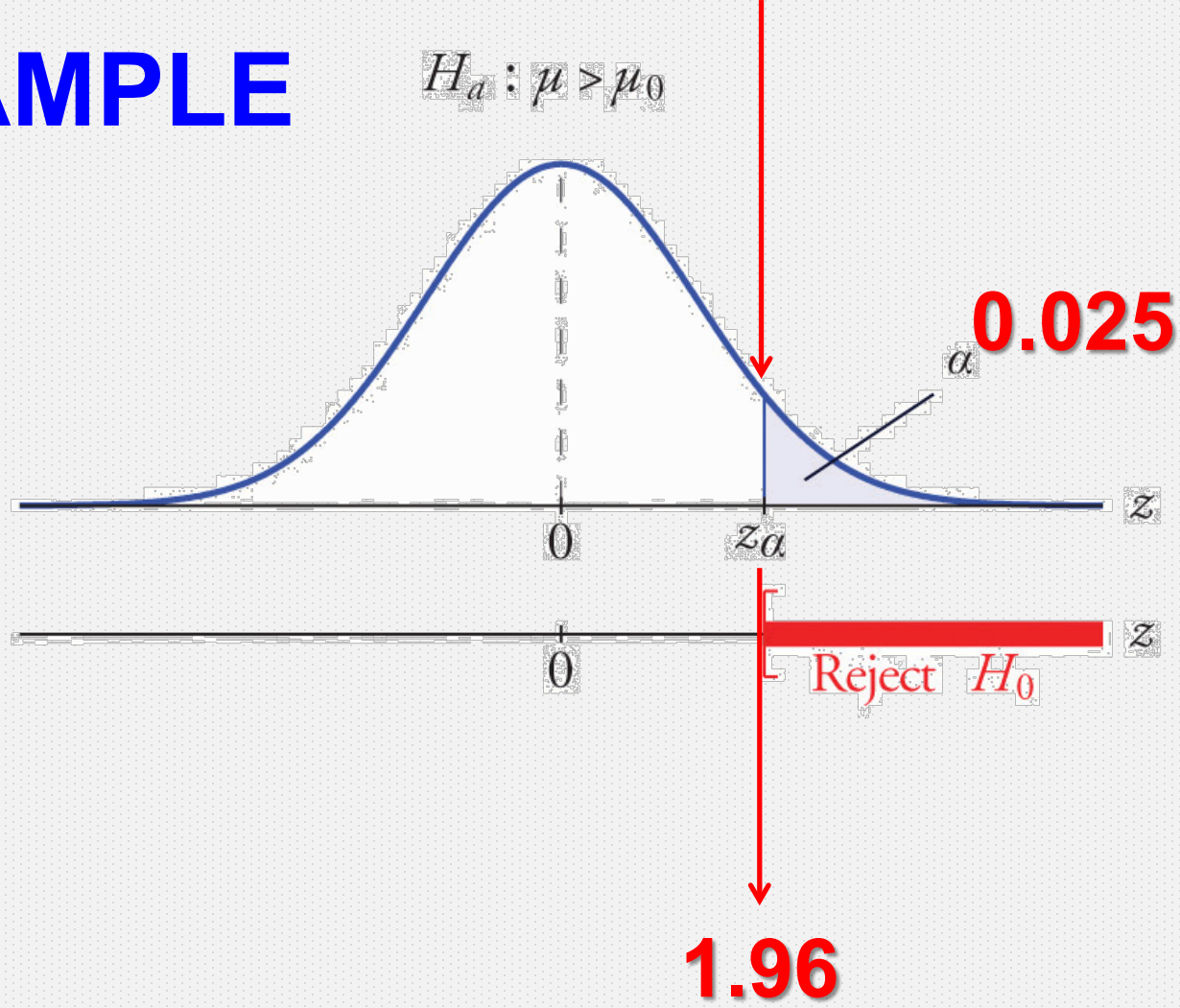


EXAMPLE

CRITICAL VALUES OF $t_{a/2}$

df	0.250	0.200	0.150	0.100	0.050	0.025	0.010	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
∞	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

EXAMPLE



$$\textit{if } Z \textit{ value is greater than } 1.96, \textit{then we will be in rejection zone of } H_o$$

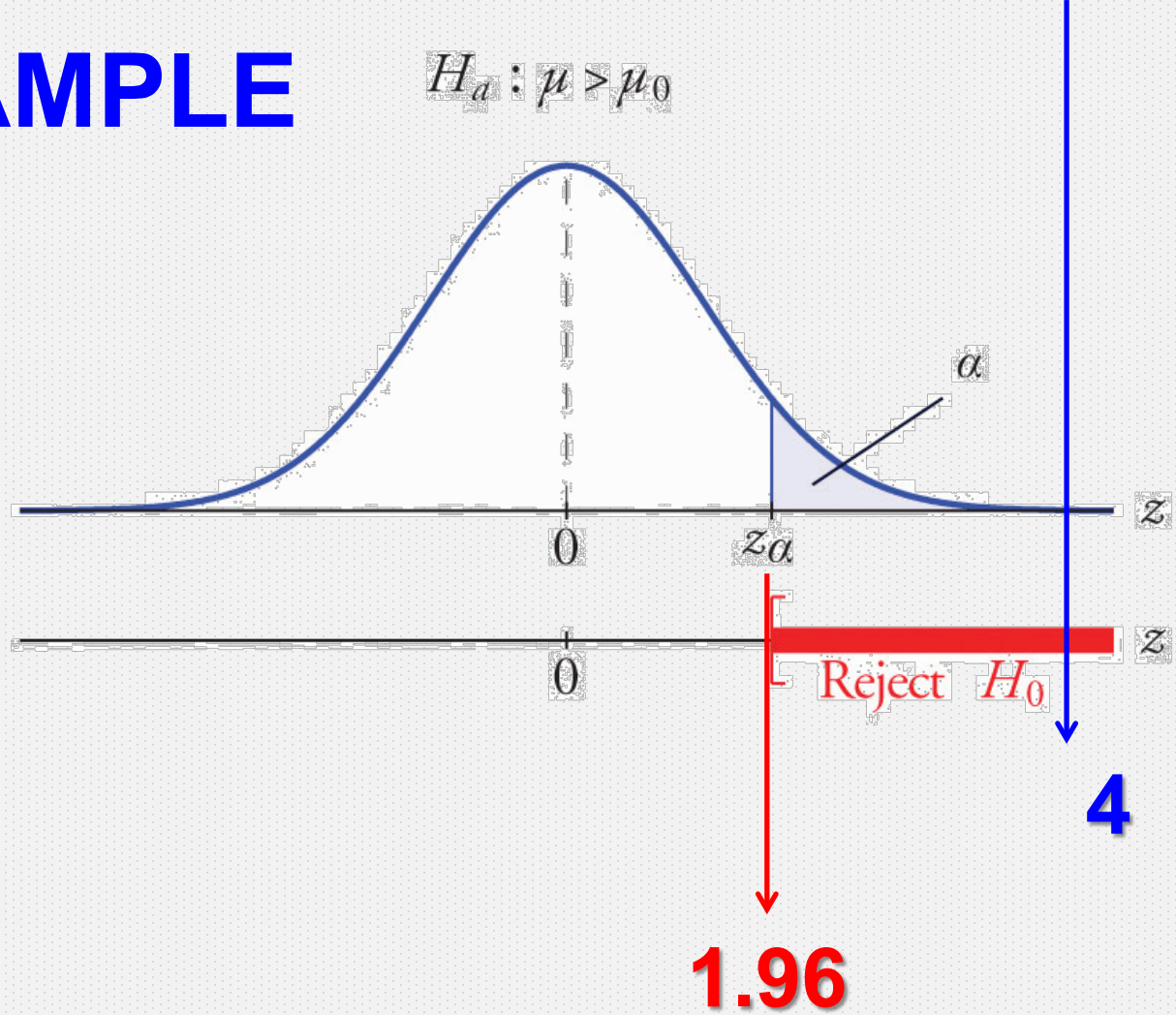
FINDING Z

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$$

Recall SAMPLING DISTRIBUTION LECTURE

$$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{17 - 15}{0.5} = \mathbf{4}$$

EXAMPLE



Z value is 4 and because it is greater than 1.96, we are in rejection zone of H_o

EXAMPLE

H_o; $\mu = 15$ The null hypothesis is rejected.

Therefore,

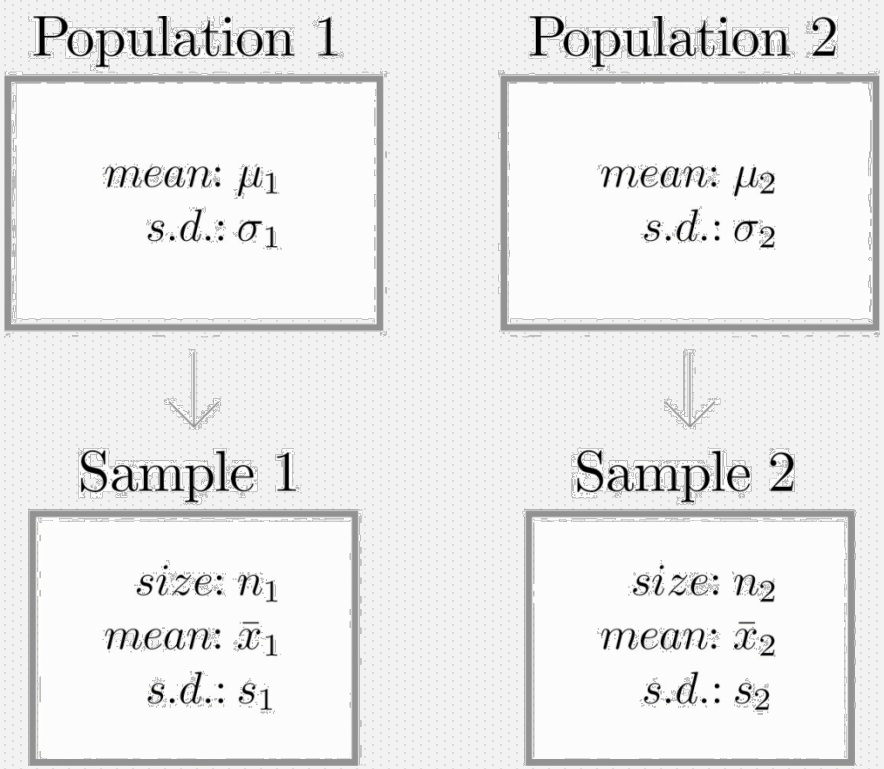
H_a; $\mu > 15$ is correct.

Thus, geologist may be right.

"A geologist claims that the average grade of an iron deposit is greater than 15 %Fe with at the 2.5% level of significance."

TWO-SAMPLE PROBLEMS

Comparison of Two Population Means: Large, **Independent** Samples



TWO-SAMPLE PROBLEMS

Samples from two distinct populations are independent if each one is drawn without reference to the other, and has no connection with the other.

The aim is to use the information in the samples to estimate the difference $\boldsymbol{\mu_1 - \mu_2}$ in the means of the two populations and to make statistically valid inferences about it.

CONFIDENT INTERVALS

Since the mean $\bar{x}_1$ of the sample drawn from Population 1 is a good estimator of μ_1 and the mean $\bar{x}_2$ of the sample drawn from Population 1 is a good estimator of μ_2, a reasonable point estimate of difference $\mu_1 - \mu_2$ is $\bar{x}_1 - \bar{x}_2$. We can widen this point estimate into a confidence interval:

$$(\bar{x}_1 - \bar{x}_2) - z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The samples must be independent, and each sample must be large $n_1 \geq 30$ and $n_2 \geq 30$

EXAMPLE

To compare satisfaction levels of two sections of course, 40 students of Section1 and 36 students of Section2 were randomly selected and were asked to rate their course on a five-point scale, with 1 being least satisfied and 5 most satisfied. The survey results are summarized in the following table:

Section1	Section2
$n_1 = 40$	$n_2 = 36$
$\bar{x}_1 = 3.81$	$\bar{x}_2 = 3.24$
$s_1 = 0.51$	$s_1 = 0.52$

EXAMPLE

Construct a point estimate and a 99% confidence interval for $\boldsymbol{\mu_1 - \mu_2}$, the difference in average satisfaction levels of students of the two sections as measured on this five-point scale.

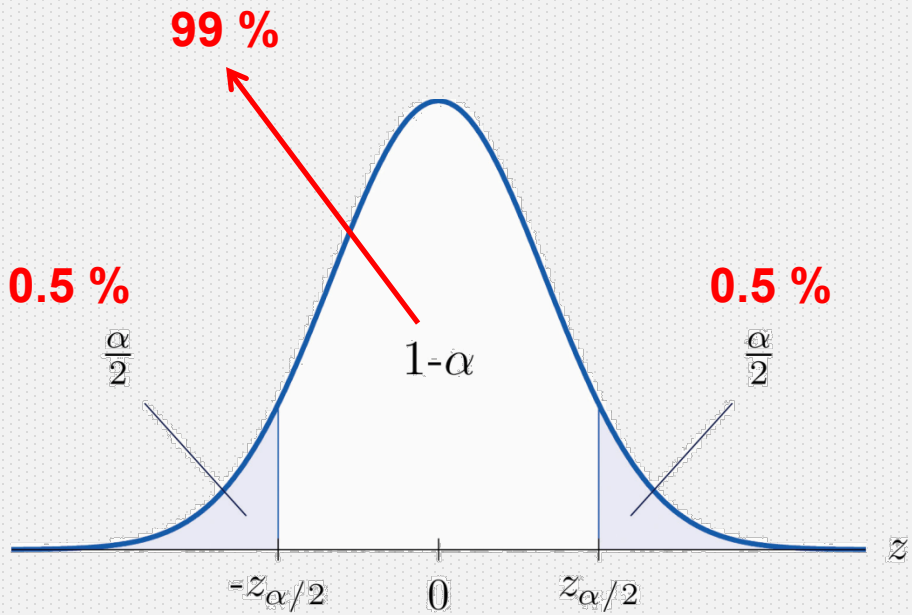
SOLUTION

To apply the formula for the CI, proceed exactly as was done in ESTIMATION LECTURE

(http://yunus.hacettepe.edu.tr/~gertunc/upload/MAD213_02NOV.pdf)

The 99 % CI means $\alpha = 1 - 0.99 = 0.01$ so $\boldsymbol{z_{\frac{\alpha}{2}}} = \boldsymbol{z_{0.005}}$ from **z** table,

99 % CONFIDENCE INTERVAL



From table we have to find $z_{0.005}$ that is

$$0.005 + 0.99 = 0.995$$

$$z_{0.005} = 2.575$$

PROBABILITY CONTENT FROM z to ∞

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

SOLUTION

$$(\bar{x}_1 - \bar{x}_2) - z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$(3.81 - 3.24) - 2.575\sqrt{\frac{0.51^2}{40} + \frac{0.52^2}{36}} < \mu_1 - \mu_2 < (3.81 - 3.24) + 2.575\sqrt{\frac{0.51^2}{40} + \frac{0.52^2}{36}}$$

$$0.57 - 0.31 < \mu_1 - \mu_2 < 0.57 + 0.31$$

$$0.26 < \mu_1 - \mu_2 < 0.88$$

Level of student satisfaction for Section1 is between 0.26 and 0.88 points higher, on this five-point scale, than that for Section2, on the basis of 99% confidence.

HYPOTHESIS TESTING

Hypotheses concerning the relative sizes of the means of two populations are tested using the same critical value procedures that were used in the case of a single population. All that is needed is to know how to express the null and alternative hypotheses and to know the formula for the standardized test statistic and the distribution that it follows.

HYPOTHESIS TESTING

The null and alternative hypotheses will always be expressed in terms of the difference of the two population means. Thus the null hypothesis will always be written

$$\mathbf{H_o; \mu_1 - \mu_2 = D_o}$$

where $\mathbf{D_o}$ is a number that is deduced from the statement of the situation.

HYPOTHESIS TESTING

$\mathbf{H_a; \mu_1 - \mu_2 < D_o}$ Left - tailed

$\mathbf{H_a; \mu_1 - \mu_2 > D_o}$ Right- tailed

$\mathbf{H_a; \mu_1 - \mu_2 \neq D_o}$ Two - tailed

STANDARDIZATION

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - \mathrm{D_o}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The samples must be independent, and each sample must be large $n_1 \geq 30$ and $n_2 \geq 30$

EXAMPLE

A laboratory experiment is conducted to compare the reaction times of hematite and magnetite to certain chemical substance. In the experiment, samples of 50 hematite and 50 magnetite are employed. The results are shown in the following table.

Hematite	Magnetite
$n_1 = 50$	$n_2 = 50$
$\bar{x}_1 = 0.36$	$\bar{x}_2 = 0.38$
$s_1^2 = 0.18$	$s_2^2 = 0.14$

EXAMPLE

Do the data present sufficient evidence to suggest that a difference between true mean reaction times for hematite and magnetite is greater than 0? Use α=0.05

Hematite	Magnetite
$n_1 = 50$	$n_2 = 50$
$\bar{x}_1 = 0.36$	$\bar{x}_2 = 0.38$
$s_1^2 = 0.18$	$s_2^2 = 0.14$

$$\mathbf{H_o; \mu_1 - \mu_2 = 0}$$
$$\mathbf{H_a; \mu_1 - \mu_2 > 0}$$

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - \mathrm{D_o}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$Z = \frac{(0.36 - 0.38) - 0}{\sqrt{\frac{0.18}{50} + \frac{0.14}{50}}} = -0.25$$

CRITICAL VALUES OF $t_{a/2}$

df	0.250	0.200	0.150	0.100	0.050	0.025	0.010	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
∞	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

$$z_{0.05} = 1.645$$

$$\mathrm{z} < z_{0.05}[-0.25 < 1.645], \mathbf{H_a}\ \boldsymbol{is\ rejected}$$

So, the answer is NO.

SMALL SAMPLE STANDARDIZATION

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - \mathrm{D_o}}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Where,

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

The samples must be independent, the populations must be normal. Small sample means that either $n_1 < 30$ or $n_2 < 30$

SMALL SAMPLE STANDARDIZATION

When sample size is less than 30, and population variance is unknown, use T-table with $df = n_1 + n_2 - 2$

DECEMBER	TOPIC
25	CORRELATION AND REGRESSION

INTRODUCTION

We can associate to each element of a population or sample two measurements x and y, particularly in the case that it is of interest to use the value of x to predict the value of y.

For example, the population could be the mining trucks, x could be the age of a particular make mining truck, and y the resale value of the truck.

INTRODUCTION

Predictor or independent variable	Response or dependent variable
Temperature in degrees Celsius	Temperature in degrees Fahrenheit
Amount spent by a research in a year	Publication made that year
Height of a 25-year-old man	Weight of the man

INTRODUCTION

Predictor or independent variable	Response or dependent variable
Temperature in degrees Celsius	Temperature in degrees Fahrenheit

Relationship between the variables is **deterministic**:

Once the value of x is known the value of y is completely determined.

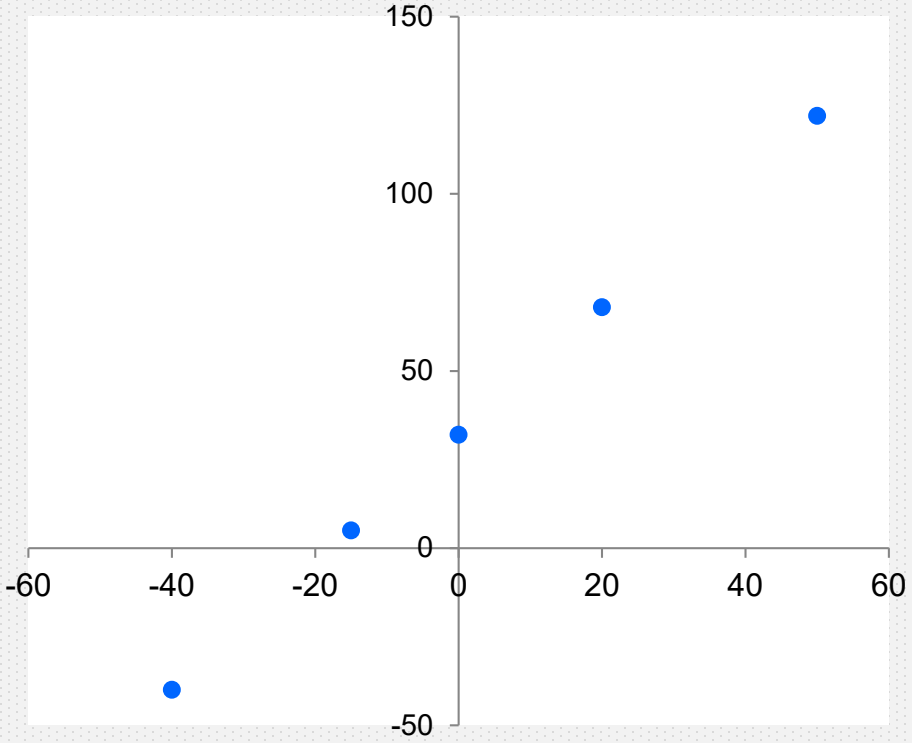
In fact there is a formula for y in terms of x:

$$y = \frac{9}{5}x + 32$$

INTRODUCTION

Choosing several values for x and computing the corresponding value for y for each one using the formula gives the table:

x	-40	-15	0	20	50
y	-40	5	32	68	122

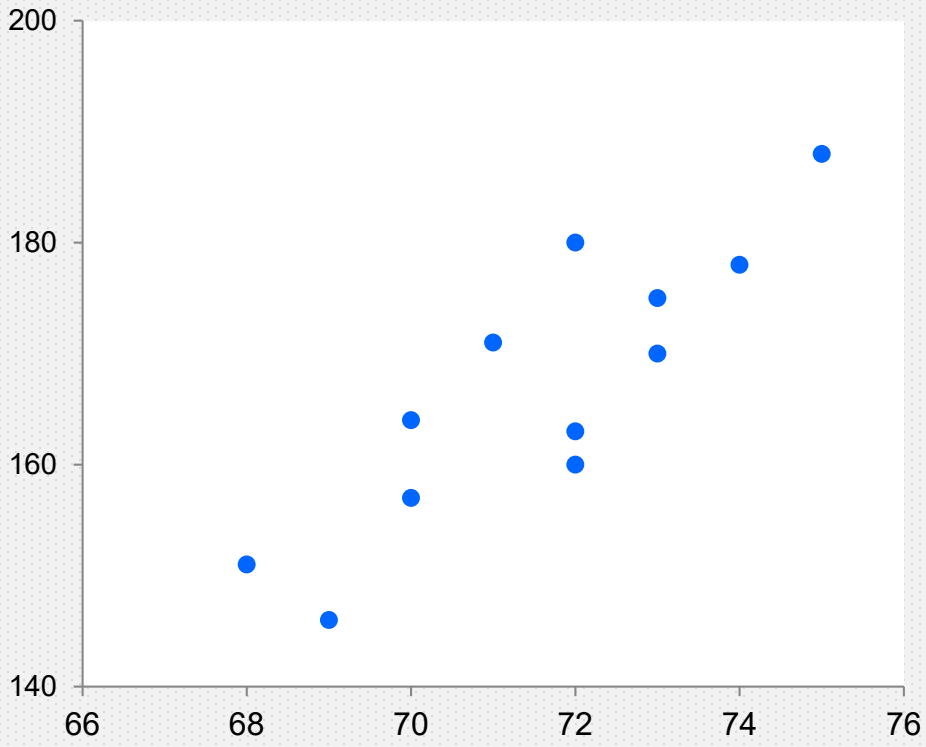


INTRODUCTION

Consider the relationship described in the last line of the table, the height x of a man aged 25 and his weight y. If we were to randomly select several 25-year-old men and measure the height and weight of each one, we might obtain a collection of (x,y) pairs something like this:

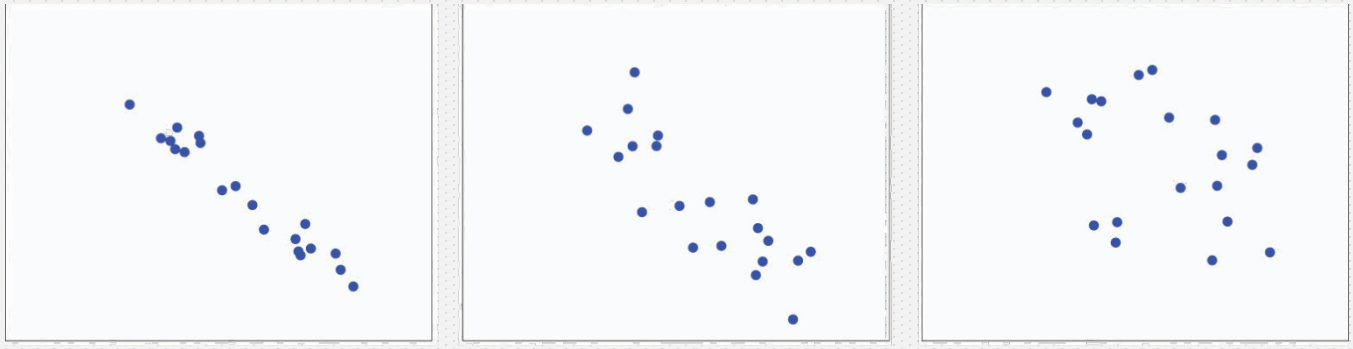
(68,151) (69,146) (70,157) (70,164) (71,171) (72,160)(72,163)(72,180)(73,170)(73,175) (74,178)(75,188)

INTRODUCTION



Such a plot is called a **scatter diagram** or **scatter plot**. Looking at the plot it is evident that there exists a linear relationship between height *x* and weight *y*, but not a perfect one. The points appear to be following a line, but not exactly. There is an element of randomness present.

LINEAR CORRELATION COEFFICIENT

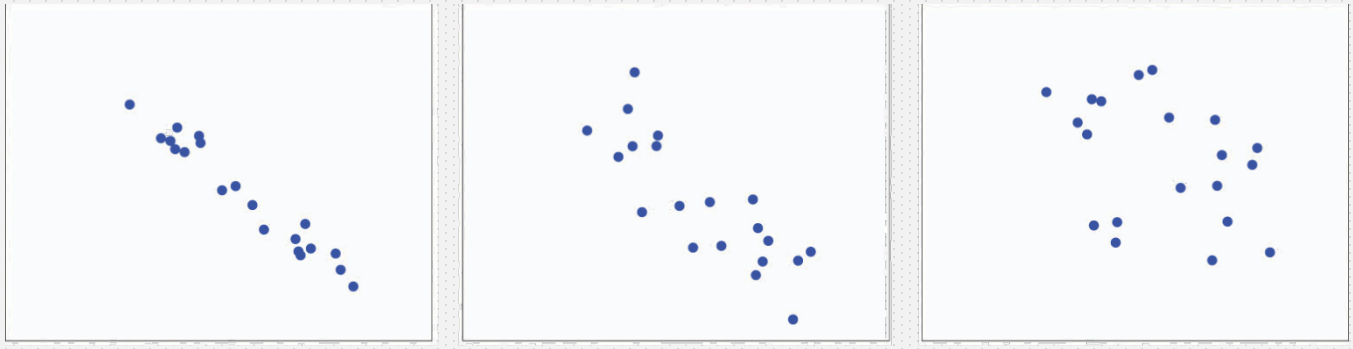


(a) (b) (c)

Linear relationships between two variables *x* and *y* of varying strengths.

It is visually apparent that in the situation in panel (a), *x* could serve as a useful predictor of *y*, it would be less useful in the situation illustrated in panel (b), and in the situation of panel (c) the linear relationship is so weak as to be practically nonexistent.

LINEAR CORRELATION COEFFICIENT



(a) (b) (c)

The *linear correlation coefficient* is a number computed directly from the data that measures the strength of the linear relationship between the two variables *x* and *y*.

LINEAR CORRELATION COEFFICIENT

$$r = \frac{ss_{xy}}{\sqrt{ss_{xx} ss_{yy}}}$$

where,

$$ss_{xx} = \sum x^2 - \frac{1}{n}\left(\sum x\right)^2$$

$$ss_{yy} = \sum y^2 - \frac{1}{n}\left(\sum y\right)^2$$

$$ss_{xy} = \sum xy - \frac{1}{n}\sum x \sum y$$

LINEAR CORRELATION COEFFICIENT

The value of r lies between -1 and 1, inclusive.

The sign of r indicates the direction of the linear relationship between x and y:

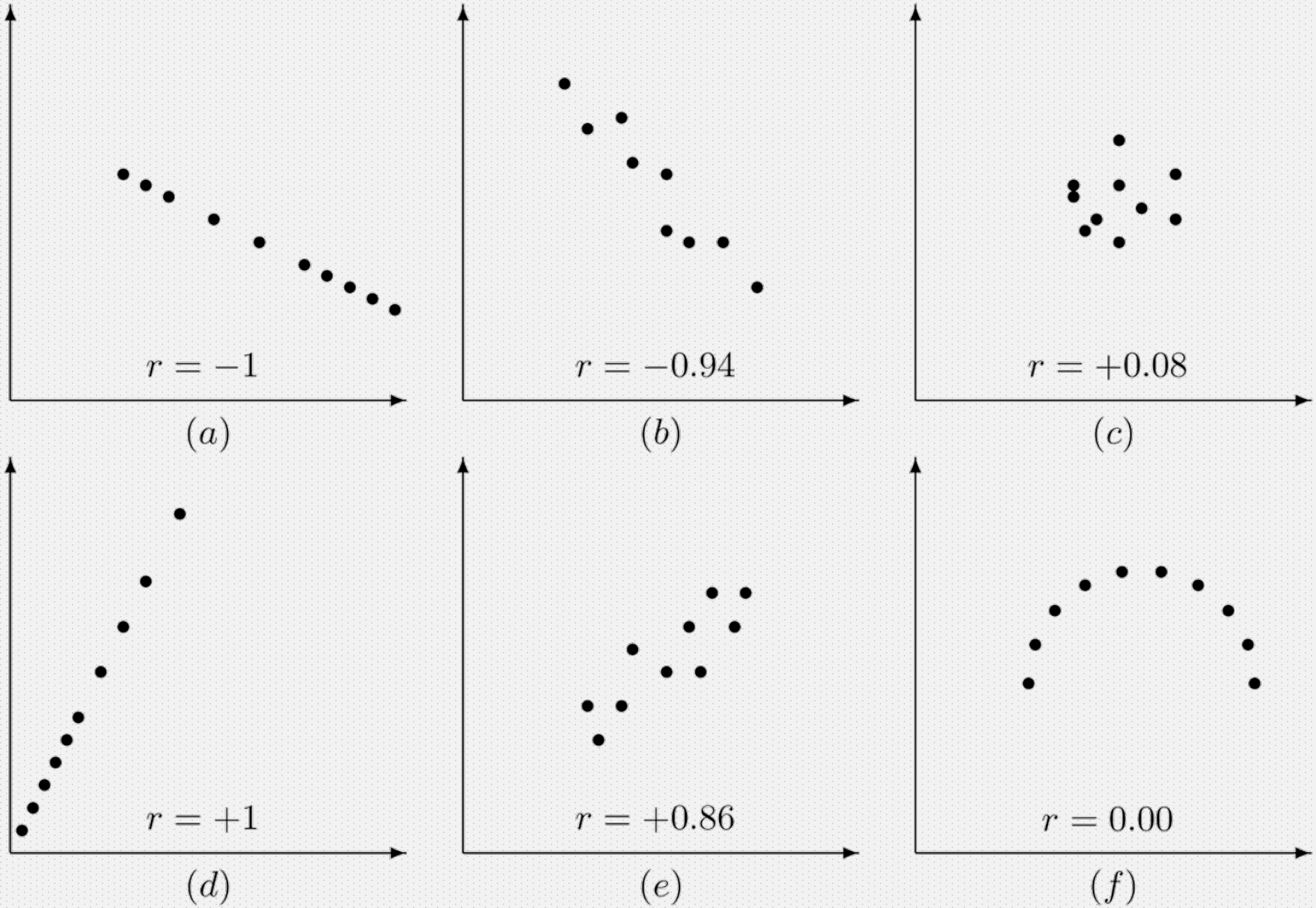
If $r < 0$ then y tends to decrease as x is increased.

If $r > 0$ then y tends to increase as x is increased.

The size of $|r|$ indicates the strength of the linear relationship between x and y:

If $|r|$ is near 1 (that is, if r is near either -1 or 1) then the linear relationship between x and y is strong.

LINEAR CORRELATION COEFFICIENT



EXAMPLE

	ppm				
	Ni	Co	$(\mathrm{Ni})^2$	$(\mathrm{Co})^2$	$\mathrm{Ni} \times \mathrm{Co}$
	23.76	11.96	564.54	143.04	284.17
	10.8	5.32	116.64	28.30	57.46
	6.68	4.68	44.62	21.90	31.26
	11.32	3.536	128.14	12.50	40.03
	6.64	3.64	44.09	13.25	24.17
	25.08	12.44	629.01	154.75	312.00
	34.72	12.92	1205.48	166.93	448.58
	31.52	13.04	993.51	170.04	411.02
	35	11.2	1225.00	125.44	392.00
	21.76	9.28	473.50	86.12	201.93
$\sum$	207.28	88.02	5424.52	922.28	2202.62

LINEAR CORRELATION COEFFICIENT

$$ss_{\mathrm{Ni}} = \sum \mathrm{Ni}^2 - \frac{1}{n}\left(\sum \mathrm{Ni}\right)^2$$

$$ss_{\mathrm{Ni}} = 5424.22 - \frac{1}{10}(207.28)^2 = 1128.03$$

$$ss_{\mathrm{Co}} = \sum \mathrm{Co}^2 - \frac{1}{n}\left(\sum \mathrm{Co}\right)^2$$

$$ss_{\mathrm{Co}} = 922.28 - \frac{1}{10}(88.02)^2 = 147.6$$

$$ss_{\mathrm{NiCo}} = \sum \mathrm{NiCo} - \frac{1}{n}\sum \mathrm{Ni} \sum \mathrm{Co}$$

$$ss_{\mathrm{Co}} = 2202.62 - \frac{1}{10}\ 207.28\ 88.02 = 378.22$$

LINEAR CORRELATION COEFFICIENT

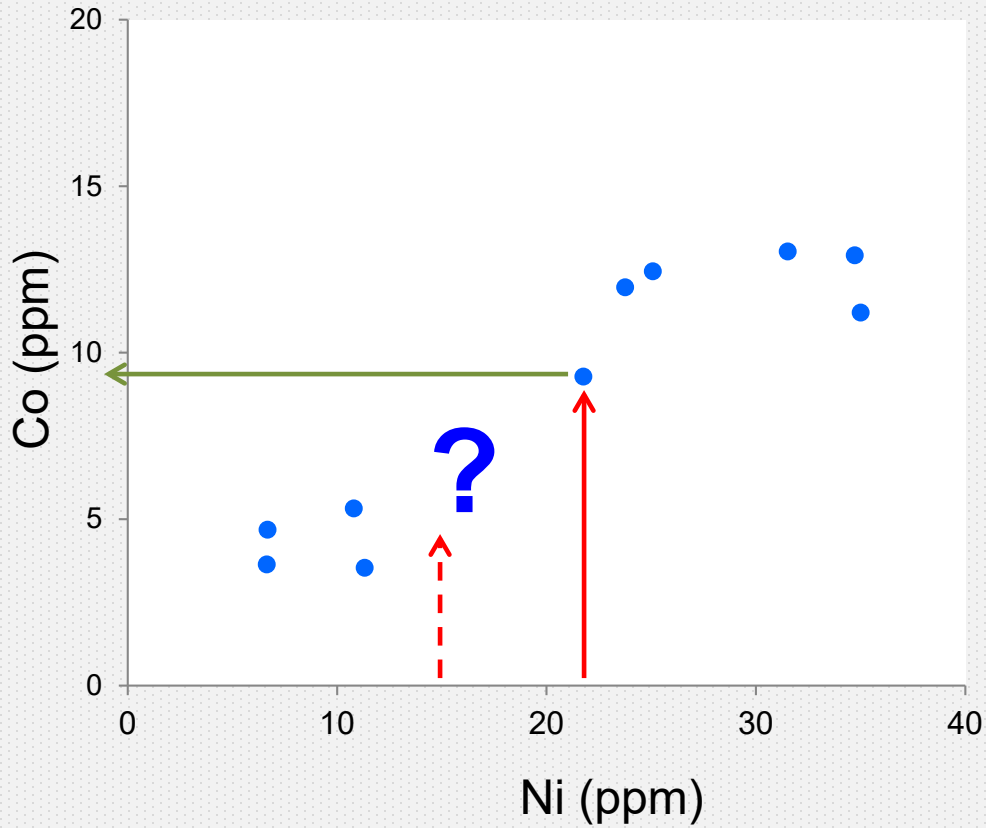
$$r = \frac{SS_{\mathrm{NiCo}}}{\sqrt{SS_{\mathrm{Ni}} SS_{\mathrm{Co}}}}$$

$$r = \frac{378.22}{\sqrt{1128.03 \times 147.6}}$$

$$r = 0.927$$

Very high correlation between Ni and Co

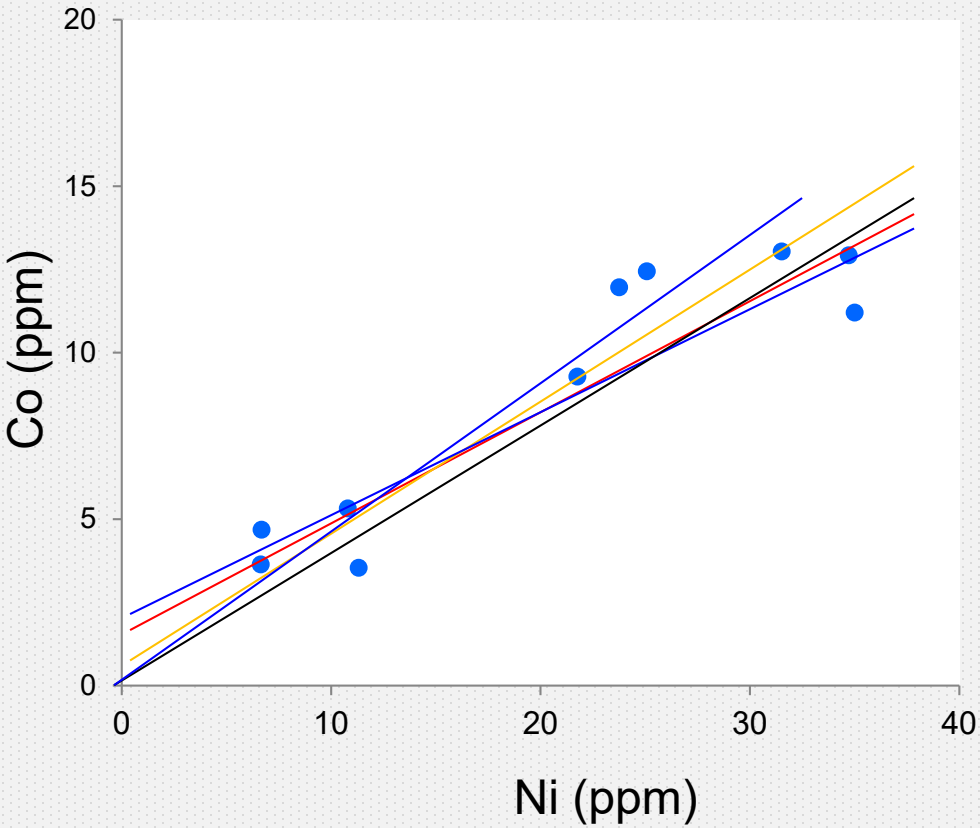
LINEAR REGRESSION



It is known that for 21.96 ppm Ni, Co value is 9.28.

What is correspondent Co value for 15 ppm Ni? Can we guess?

LINEAR REGRESSION



Which line suits best?

Each possible linear line has an equation in the form of:

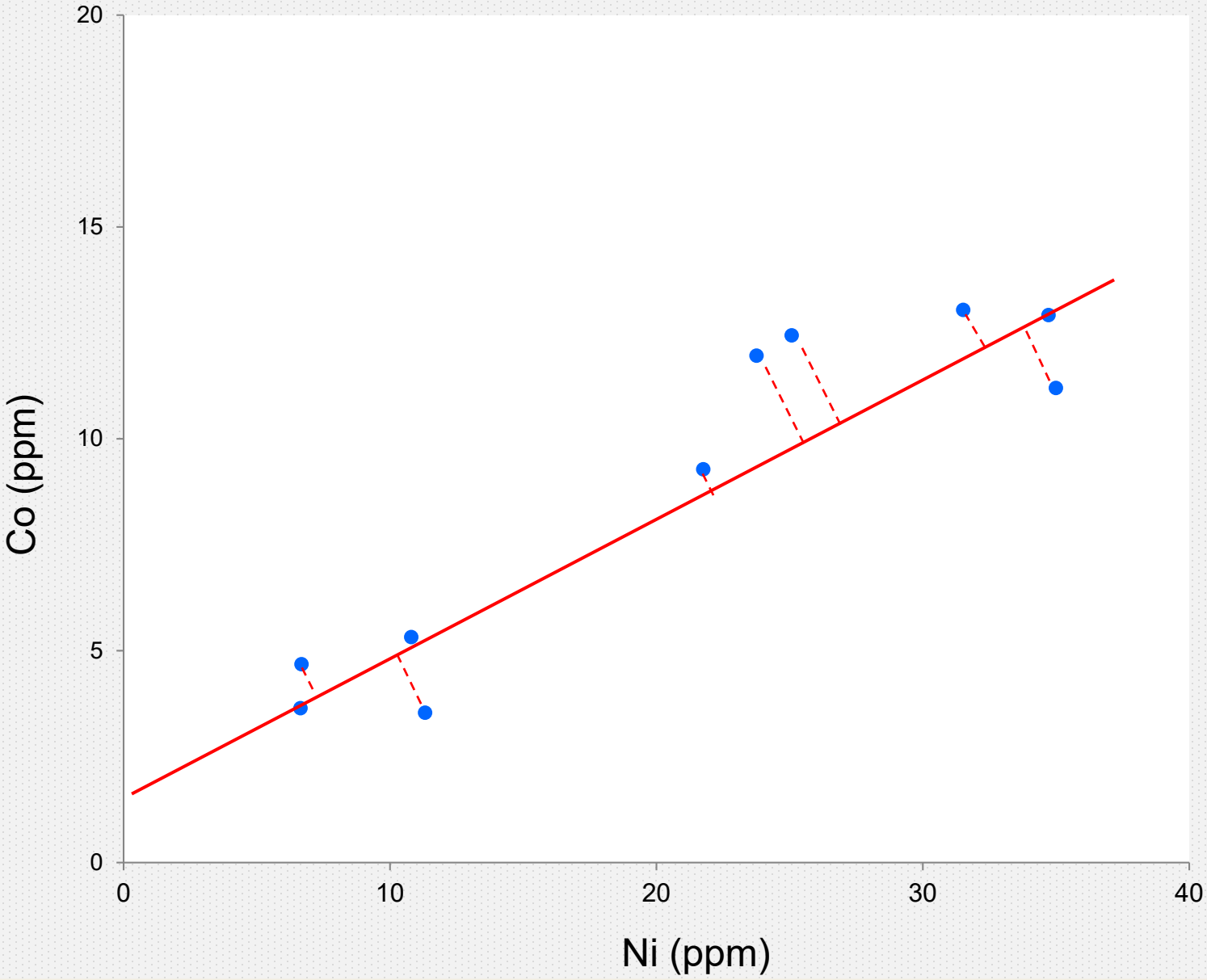
$$y = \beta_0 + \beta_1 x$$

This model is known to be deterministic.

For example, whenever $x = 20$, $y = \beta_0 + 20\beta_1$

(Remember Celsius – Fahrenheit relation)

LINEAR REGRESSION



LINEAR REGRESSION

Unlike the **deterministic model**, statistics use **probabilistic models**.

$$y = \beta_0 + \beta_1 x + \varepsilon$$

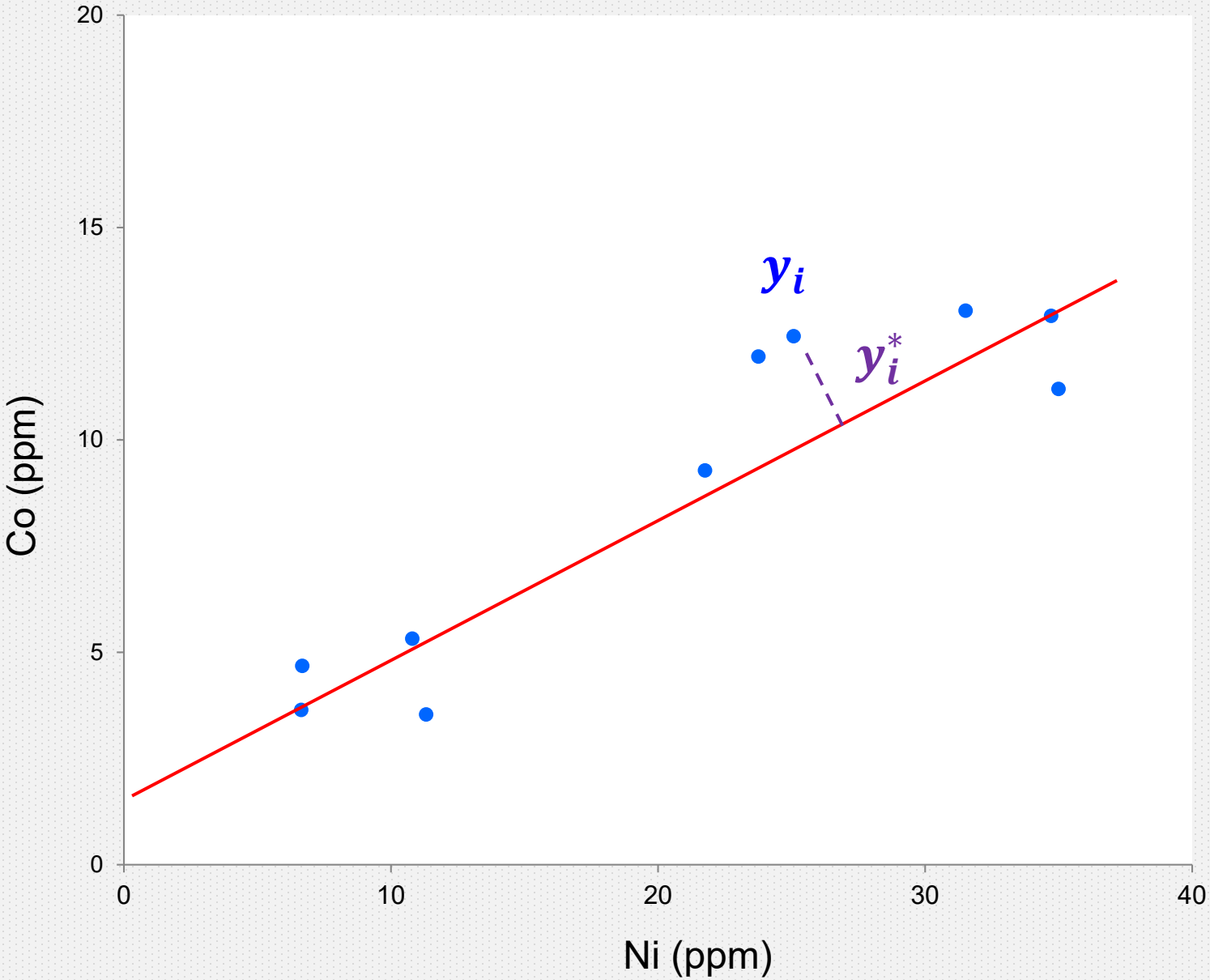
where, ε is a random variable with zero mean.

With each chosen parameter β_0 and β_1 there is different linear line. In each candidate line there will be some amount of error. That is ε.

So β_0 and β_1 estimaters denoted as follows.

$$y^* = \beta_0^* + \beta_1^* x$$

LINEAR REGRESSION



LINEAR REGRESSION

$$y_i - y_i^* = \boldsymbol{error}$$

For the sake of simplicity and mathematical handling, error is squared in statistics. One of aims in statistics is to determine minimum error.

Derivative of the function (which is in second order because of ^2) will gives minimum value of error.

LINEAR REGRESSION

$$\boldsymbol{error^2} = (y_i - y_i^*)^2$$

$\sum_i^n \boldsymbol{error^2}$: Sum of squared errors (SSE)

$$\sum_i^n \boldsymbol{error^2} = \sum_i^n (y_i - (\beta_0^* - \beta_1^* x_i))^2$$

Minimizing SSE by taking partial derivatives according to β_0^* and β_1^*

PARTIAL DERIVATIVE β_0^*

$$\frac{\delta \mathrm{SSE}}{\delta \beta_0^*} = \frac{\delta[\sum_i^n (y_i - (\beta_0^* - \beta_1^* x_i))^2 \ ]}{\delta \beta_0^*}$$

$$\frac{\delta \mathrm{SSE}}{\delta \beta_0^*} = -\sum_i^n 2[y_i - (\beta_0^* - \beta_1^* x_i)]$$

$$\frac{\delta \mathrm{SSE}}{\delta \beta_0^*} = -2\left[\sum_i^n y_i - n\beta_0^* - \beta_1^* \sum_i^n x_i\right]$$

PARTIAL DERIVATIVE β_1^*

$$\frac{\delta \mathrm{SSE}}{\delta \beta_1^*} = \frac{\delta[\sum_i^n (y_i - (\beta_0^* - \beta_1^* x_i))^2 \ ]}{\delta \beta_1^*}$$

$$\frac{\delta \mathrm{SSE}}{\delta \beta_1^*} = -\sum_i^n 2[y_i - (\beta_0^* - \beta_1^* x_i)]\, x_i$$

$$\frac{\delta \mathrm{SSE}}{\delta \beta_1^*} = -2\left[\sum_i^n x_i y_i - \beta_0^* \sum_i^n x_i - \beta_1^* \sum_i^n x_i^2\right]$$

MINIMIZING SSE

$$\frac{\delta \mathrm{SSE}}{\delta \beta_1^*} = -2\left[\sum_i^n x_i y_i - \beta_0^* \sum_i^n x_i - \beta_1^* \sum_i^n x_i^2\right] = 0$$

We obtain β_1^*,

$$\beta_1^* = \frac{n\sum_i^n x_i y_i - \sum_i^n x_i \sum_i^n y_i}{n\sum_i^n {x_i}^2 - (\sum_i^n x_i)^2}$$

MINIMIZING SSE

$$\frac{\delta \mathrm{SSE}}{\delta \beta_0^*} = -2\left[\sum_i^n y_i - n\beta_0^* - \beta_1^* \sum_i^n x_i\right] = 0$$

We obtain β_0^*,

$$\beta_0^* = \bar{y} - \beta_1^* \bar{x}$$

EXAMPLE

We have the following data and we want to fit a line to the experimental values.

	ppm				
	Ni	Co	$(\mathrm{Ni})^2$	$(\mathrm{Co})^2$	$\mathrm{Ni} \times \mathrm{Co}$
	23.76	11.96	564.54	143.04	284.17
	10.8	5.32	116.64	28.30	57.46
	6.68	4.68	44.62	21.90	31.26
	11.32	3.536	128.14	12.50	40.03
	6.64	3.64	44.09	13.25	24.17
	25.08	12.44	629.01	154.75	312.00
	34.72	12.92	1205.48	166.93	448.58
	31.52	13.04	993.51	170.04	411.02
	35	11.2	1225.00	125.44	392.00
	21.76	9.28	473.50	86.12	201.93
$\sum$	207.28	88.02	5424.52	922.28	2202.62

EXAMPLE

$$\beta_1^* = \frac{10\sum_i^{10} \mathrm{Ni}_i \mathrm{Co}_i - \sum_i^{10} \mathrm{Ni}_i \sum_i^{10} \mathrm{Co}_i}{10\sum_i^{10} \mathrm{Ni}_i{}^2 - (\sum_i^{10} \mathrm{Ni}_i)^2}$$

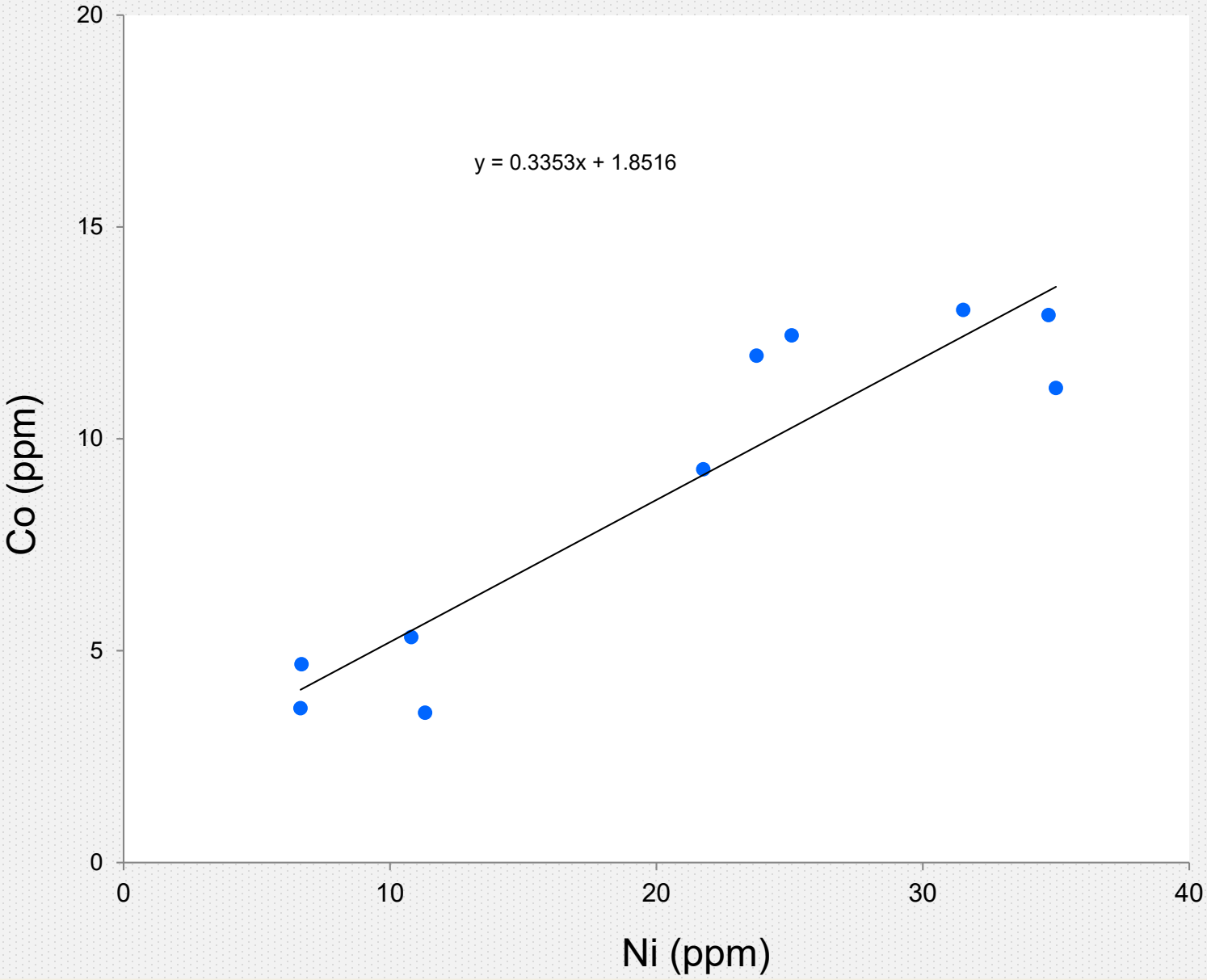
$$\beta_1^* = \frac{10 \times 2202.62 - 207.28 \times 88.02}{10 \times 5424.52 - (207.28)^2}$$

$$\beta_1^* = 0.335$$

EXAMPLE

$$\beta_0^* = \overline{\mathrm{Co}} - \beta_1^* \overline{\mathrm{Ni}}$$

$$\beta_0^* = 8.80 - 0.335 \times 20.73$$
$$\beta_0^* = 1.852$$

$$y = 0.335x + 1.852$$

EXAMPLE